



Modelling Financial Markets

Chapter 14: Deep Learning and Time-Series Foundation Models



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Acronyms used in this material

- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CPU: Central Processing Unit
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: Expected Shortfall
- ▣ FHS: Filtered Historical Simulation
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GIFT: General Time Series Forecasting Model Evaluation (benchmark GIFT-Eval)
- ▣ GRU: Gated Recurrent Unit
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ LLM: Large Language Model
- ▣ LSTM: Long Short-Term Memory (network)
- ▣ MCS: Model Confidence Set
- ▣ MLP: MultiLayer Perceptron
- ▣ MSE: Mean Squared Error
- ▣ QLIKE: Quasi-LIKElihood (loss function)
- ▣ RNN: Recurrent Neural Network
- ▣ RV: Realised Variance
- ▣ US: United States

Outline

Why Deep Learning for Financial Time Series?

From the Perceptron to Deep Networks

Recurrent Networks and the LSTM

Attention and Transformers

Time-Series Foundation Models

Test 1: Forecasting Returns

Test 2: Forecasting Realised Volatility

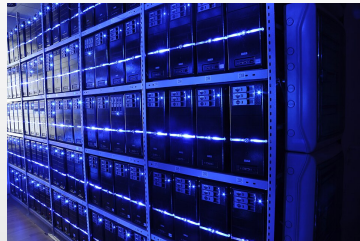
Test 3: VaR 1% and ES 2.5%

Practice and Conclusions

References

The Promise and the Question

- **Deep learning:** models built from many layers of simple non-linear units, trained on data
 - ▶ It transformed image recognition and language processing
 - ▶ Since 2024: **foundation models** for time series, pretrained on billions of observations
- The question of this chapter
 - ▶ Do these models forecast **returns**, **volatility** and **tail risk** better than the classical models of Chapters 5–9?
 - ▶ We answer with the evaluation tools of Chapter 8



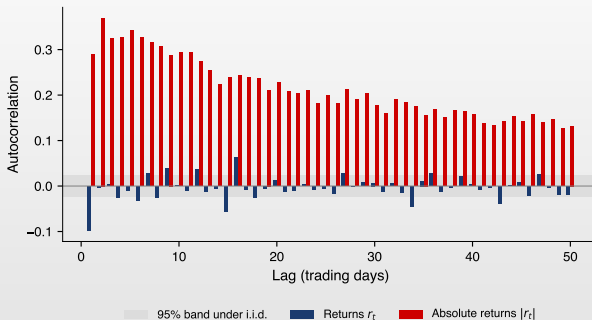
A data centre: deep networks are trained on racks of such servers

Photo: [BalticServers.com](#) (2013); CC BY-SA 3.0; [Wikimedia Commons](#)

Three Forecasting Targets

- ▣ **Returns** r_t (conditional mean)
 - ▶ Very low signal-to-noise ratio; predictable patterns are traded away (Chapter 2)
- ▣ **Volatility** (conditional variance)
 - ▶ Persistent and clustered (Chapters 1 and 5): there is a signal to learn
- ▣ **Tail quantiles**: VaR 1% and ES 2.5%
 - ▶ Need the shape of the lower tail, not just the centre of the distribution (Chapter 7)
- ▣ Each target gets its own loss function and its own benchmark model

Where Is the Signal?



- S&P 500 daily returns since 2000 (6 718 days): autocorrelation at lags 1 to 50
- Grey band: 95% range under independence, $\pm 1.96/\sqrt{T} = \pm 0.024$

Interpretation: Mean versus Size of Moves

- ▣ Returns: first-order autocorrelation -0.099
 - ▶ 15 of 50 lags outside the band, but small and of changing sign: little to learn about the mean
- ▣ Absolute returns: autocorrelation 0.29 at lag 1, still 0.13 at lag 50
 - ▶ All 50 lags outside the band: volatility clustering with long memory
- ▣ Prediction for the rest of the chapter: any flexible model can find the volatility signal; none should find much in the mean

What We Do in This Chapter

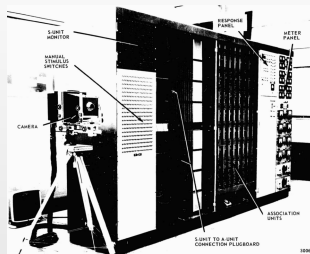
- ▣ From the perceptron to deep networks: neurons, layers, training
- ▣ Recurrent networks and the LSTM: why memory is hard to learn
- ▣ Attention and Transformers: patches, positions, quantile outputs
- ▣ Foundation models: Chronos, TimesFM, Moirai; zero-shot use and pretraining leakage
- ▣ Three empirical tests: returns, realised volatility, VaR 1% and ES 2.5%

Section Summary

- ▣ Three targets with very different signal-to-noise ratios: mean, volatility, tail
- ▣ Absolute returns are strongly autocorrelated, returns are not
- ▣ Every new model is judged against a simple benchmark with the tests of Chapter 8

The Perceptron (1958)

- **Artificial neuron:** weighted sum of inputs plus a bias, passed through an activation function
 - ▶ $a = \varphi(w^T x + b)$
 - ▶ w : weights, b : bias, φ : activation (non-linear function)
- **Rosenblatt (1958):** the perceptron learns w from labelled examples
 - ▶ Built in hardware as the Mark I Perceptron (photo)
 - ▶ One layer only separates classes with a straight line (hyperplane)



The Mark I Perceptron, Cornell Aeronautical Laboratory

Photo: J.C. Hay, A.E. Murray (1960); public domain; [Wikimedia](#)

[Commons](#)

The Multilayer Perceptron (MLP)

- Layers applied one after another: $h^{(\ell)} = \varphi(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$, $h^{(0)} = x$
 - ▶ $h^{(\ell)}$: hidden state of layer ℓ ; the last layer is linear for regression
- Common activations
 - ▶ ReLU (rectified linear unit): $\max(0, z)$
 - ▶ $\tanh(z) \in (-1, 1)$; sigmoid $\sigma(z) = 1/(1 + e^{-z}) \in (0, 1)$
- **Universal approximation** (Hornik, Stinchcombe & White (1989)): one hidden layer with enough units approximates any continuous function
 - ▶ It says nothing about how much data is needed: the key issue in finance

Training a Network

- ▣ Loss: mean squared error (MSE) $\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$ for point forecasts; pinball loss for quantiles
- ▣ **Backpropagation** (Rumelhart, Hinton & Williams (1986)): the chain rule gives the gradient of the loss with respect to every weight
- ▣ **Gradient descent**: $\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L}$, learning rate η
 - ▶ Mini-batches: gradient on a random subset of examples; **epoch**: one pass through the data
 - ▶ Adam (Kingma & Ba (2015)): step sizes adapted per parameter from running moments of the gradient

Overfitting and How Deep Learning Fights It

- ▣ Networks have many parameters relative to financial samples
 - ▶ Our small LSTM has 1 233 parameters; Chronos-2 has about 119 million
- ▣ **Early stopping**: stop training when the loss on a validation set stops falling
 - ▶ In time series the validation set must come after the training set (Chapter 13: purged cross-validation)
- ▣ **Weight decay**: penalty $\lambda \|\theta\|^2$; **dropout**: random switching-off of units during training
- ▣ **Ensembles**: average several networks trained from different random seeds

Deep Learning Comes of Age

- ▣ [LeCun, Bengio & Hinton \(2015\)](#): review of the methods behind the breakthroughs in vision and speech
- ▣ 2024: Nobel Prize in Physics to Hopfield and Hinton for foundational work on neural networks
- ▣ In finance: [Gu, Kelly & Xiu \(2020\)](#) find that neural networks improve monthly stock-return predictions through non-linear interactions (Chapter 13)
 - ▶ Cross-section of thousands of stocks, not one index: much more data



Geoffrey Hinton, Nobel laureate 2024

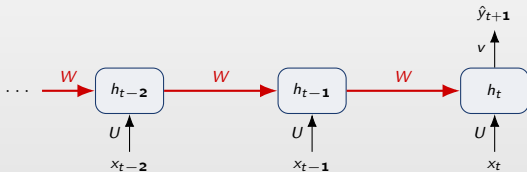
Photo: [Arthur Petron \(2024\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

Section Summary

- Neuron: weighted sum plus non-linear activation; MLP: layers of neurons
- Training: backpropagation plus gradient descent (Adam) on mini-batches
- Flexibility is cheap, data are not: early stopping, weight decay and ensembles are essential

Recurrent Neural Networks (RNN)

- A hidden state carries information forward in time
 - ▶ $h_t = \tanh(W h_{t-1} + U x_t + b)$, $\hat{y}_{t+1} = v^\top h_t + c$
 - ▶ The same weights W, U, b at every step: the number of parameters does not grow with the sequence length
- Unrolled over time, an RNN is a very deep network with shared weights



Why Long Memory Is Hard: the Vanishing Gradient

- Backpropagation through time multiplies one Jacobian matrix per step

- ▶ $\frac{\partial h_T}{\partial h_{T-k}} = \prod_{j=0}^{k-1} \text{diag}(1 - h_{T-j}^2) W$

- ▶ Scalar case: $|w| \tanh'(\cdot) \leq |w|$, so the gradient is at most $|w|^k$

- With $|w| = 0.9$: $0.9^{20} = 0.12$, $0.9^{50} = 0.005$

- ▶ Bengio, Simard & Frasconi (1994): gradient-based learning of long-range dependencies is difficult
- ▶ Exploding gradients (norms above 1) are the mirror problem; clipping the gradient norm is the usual fix

The LSTM Cell

- **Long short-term memory** (Hochreiter & Schmidhuber (1997)): a cell state c_t with gates that decide what to keep, write and output
 - ▶ Forget gate: $f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$
 - ▶ Input gate: $i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$; candidate $g_t = \tanh(W_g x_t + U_g h_{t-1} + b_g)$
 - ▶ Output gate: $o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$
- Updates: $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, $h_t = o_t \odot \tanh(c_t)$
 - ▶ $\odot$: element-by-element product; gates take values in $(0, 1)$

Why the LSTM Remembers

- Along the cell state the only multiplier is the forget gate
 - ▶ $\frac{\partial c_T}{\partial c_{T-k}} = \prod_{j=0}^{k-1} f_{T-j}$ (direct path)
 - ▶ With $f \approx 1$ the gradient survives many steps
- Constant forget gate $f = \sigma(b_f)$
 - ▶ $b_f = 0$: $f = 0.5$, $0.5^{50} \approx 9 \times 10^{-16}$
 - ▶ $b_f = 3$: $f = 0.953$, $0.953^{50} \approx 0.09$; half-life 14 steps
- GRU (gated recurrent unit): a lighter variant with two gates

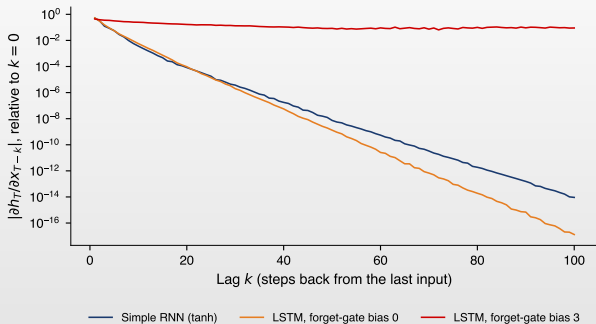


Sepp Hochreiter, co-inventor of the LSTM

Photo: Eulenreich (2015); CC BY-SA 4.0; Wikimedia

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The Vanishing Gradient, Measured



- ▣ $|\partial h_T / \partial x_{T-k}|$ by automatic differentiation; 16 hidden units, 50 random initialisations, log scale
- ▣ The same random inputs for the three networks; only the architecture and the forget-gate bias change

Interpretation: Architecture Is Not Enough

- Simple RNN: gradient share 8.3×10^{-5} at lag 20 and 7.4×10^{-9} at lag 50
 - ▶ Practically nothing is learned from inputs more than a few weeks back
- LSTM with forget-gate bias 0: 1.3×10^{-9} at lag 50, no better than the RNN
 - ▶ At initialisation $f \approx 0.5$: the gate halves the memory at every step
- LSTM with forget-gate bias 3: 8.4% at lag 50, 9.0% at lag 100
 - ▶ Matches the constant-gate calculation $0.953^{50} \approx 0.09$
 - ▶ Practical rule: initialise the forget-gate bias at a positive value

How Big Is a Small LSTM?

- Parameters of one LSTM layer with input size d and hidden size h
 - ▶ $4 \times (hd + h^2 + h)$ (PyTorch uses two bias vectors: $4 \times (hd + h^2 + 2h)$)
 - ▶ Four blocks: three gates and the candidate
- Our return model: $d = 1$, $h = 16$, linear output: 1 233 parameters
 - ▶ Trained on at most 4,000 daily returns: about three observations per parameter
- Application in finance: [Fischer & Krauss \(2018\)](#) use LSTMs to predict whether S&P 500 constituents beat the cross-sectional median the next day

Section Summary

- RNN: one hidden state, shared weights; long memory fails because gradients vanish
- LSTM: additive cell state and gates; the forget gate controls how long information lives
- Even small LSTMs have thousands of parameters: a lot for one financial series

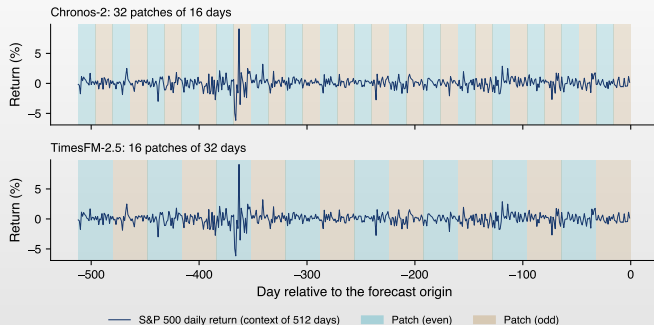
Attention: Looking Back Directly

- Each time step asks a question (query) and compares it with every past step (key)
 - ▶ $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$ (Vaswani et al. (2017))
 - ▶ $Q = XW_Q$, $K = XW_K$, $V = XW_V$: linear projections of the inputs X ; d_k : key dimension
 - ▶ softmax turns scores into weights that sum to one; division by $\sqrt{d_k}$ keeps the scores from saturating it
- No product of Jacobians over time: any past step is one hop away
 - ▶ Cost grows with the square of the number of inputs n : $O(n^2)$

Positions and Patches

- **Positional encoding:** attention treats its inputs as a set, so the order must be added
 - ▶ Sinusoids of different frequencies added to the inputs (original Transformer)
 - ▶ Rotary embeddings: rotate queries and keys by an angle proportional to the position (used by Chronos-2)
- **Patching** (Nie et al. (2023)): a token is a block of consecutive observations, not a single one
 - ▶ 512 daily returns in patches of 16 days: 32 tokens instead of 512, about 256 times fewer attention scores
 - ▶ Each token summarises local dynamics (a few weeks of market)

Patching a 512-Day Context



- The last 512 daily returns of the S&P 500, split as Chronos-2 (16 days) and TimesFM-2.5 (32 days) read them
- Each shaded block becomes one embedding vector for the Transformer

Other Neural Architectures for Forecasting

- DeepAR ([Salinas et al. \(2020\)](#)): an RNN that outputs the parameters of a predictive distribution, trained on many related series
- N-BEATS ([Oreshkin et al. \(2020\)](#)): stacks of fully connected blocks that forecast and subtract, without recurrence or attention
- PatchTST ([Nie et al. \(2023\)](#)): Transformer on patches, each series processed separately
- Common thread: learn across many series, because one financial series is short
 - ▶ Foundation models push this idea to its limit: learn across all available series

Section Summary

- ▣ Attention compares every step with every other step: no vanishing gradient through time, but quadratic cost
- ▣ Positions must be encoded; patches reduce the number of tokens and carry local context
- ▣ Modern forecasting networks learn across many series

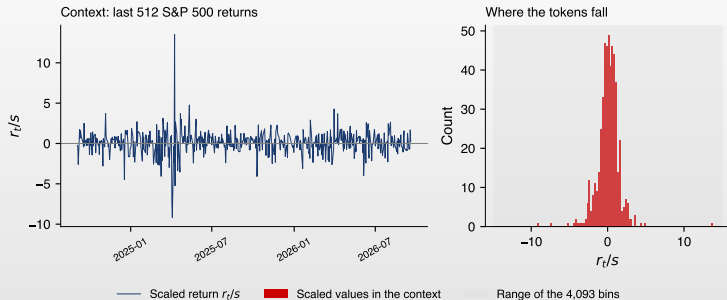
What Is a Foundation Model?

- A large network pretrained once on a very broad corpus, then used on new tasks
 - ▶ Idea borrowed from large language models (LLM): pretrain on text, then use on any question
- **Zero-shot** forecasting: no training on the target series; only its recent history (the **context**) is passed at forecast time
 - ▶ Output: a set of quantiles of the future values, not only a point
 - ▶ Fine-tuning (further training on the target series) is possible but not used here
- Appeal for finance: one model for thousands of series, no estimation per series

Chronos: Time Series as a Language

- ▣ Ansari et al. (2024): turn values into tokens and train a language model (T5) with cross-entropy
 - ▶ Mean scaling: divide the context by $s = \frac{1}{C} \sum_t |x_t|$
 - ▶ Quantisation: 4,093 equally spaced bin centres in $[-15, 15]$ (width 0.0073), plus special tokens
 - ▶ Forecast: sample future tokens, map back to values, read quantiles from the samples
- ▣ Pretraining corpus: public datasets (energy, transport, retail, weather, web, finance and economics) plus synthetic series (TSMixup, KernelSynth)
 - ▶ Model sizes 20 to 710 million parameters; the small version has 46 million

What Chronos Sees: Scaled Returns and Tokens



- Context: the last 512 S&P 500 daily returns; scale $s = 0.672\%$ (mean absolute return)
- Right: where the scaled values fall within the token range $[-15, 15]$

Interpretation: a Vocabulary Mostly Unused

- ▣ Scaled returns range from -9.2 to 13.5; 96% lie within ± 3
 - ▶ The 512 returns use 336 distinct tokens out of 4,093; ± 3 alone spans 818 bins
- ▣ Scaling makes the model free of units: 1% in a calm year and 3% in a crisis can map to the same token
 - ▶ Good for transfer across series; the level of volatility must be recovered from the context itself
- ▣ A crash day inflates s and compresses all other days towards zero (Seminar, A6)

Chronos-Bolt and Chronos-2

- ▣ **Chronos-Bolt:** patches of 16 values and a direct quantile output instead of token sampling
 - ▶ Quantile levels 10%, 20%, ..., 90% only; small version 48 million parameters
- ▣ **Chronos-2** ([Ansari et al. \(2025\)](#)): robust scaling, patches of 16, attention across time and across groups of related series
 - ▶ 21 quantile levels: 1%, 5%, 10%, ..., 95%, 99%; about 119 million parameters
 - ▶ Training corpus: selected datasets of the Chronos and [GIFT-Eval](#) corpora plus synthetic multivariate series
 - ▶ Benchmarks: [GIFT-Eval](#) ([Aksu et al. \(2024\)](#)) and fev-bench

TimesFM, Moirai and Lag-Llama

- **TimesFM** ([Das et al. \(2024\)](#)): decoder-only Transformer on patches of 32 values
 - ▶ Pretrained on about 10^{11} time points: Google Trends, Wikipedia page views, synthetic series, M4, electricity, traffic, weather
 - ▶ TimesFM-2.5: mean plus deciles; the released checkpoint has about 231 million parameters
- **Moirai** ([Woo et al. \(2024\)](#)): any number of variables, mixture-distribution output; Moirai 2.0 ([Liu et al. \(2025\)](#))
 - ▶ Released under a non-commercial licence; not used in our tests
- **Lag-Llama** ([Rasul et al. \(2023\)](#)): decoder-only model with lagged values as features and a Student- t output

The Three Models We Test

Model	Parameters	Patch	Quantile levels	Weights published
Chronos-Bolt (small)	48 M	16	10%–90%	25.11.2024
TimesFM-2.5	231 M	32	mean, 10%–90%	01.10.2025
Chronos-2	119 M	16	1%, 5%, . . . , 99%	30.10.2025

- All three run on an ordinary CPU; context of 512 daily observations, one-day-ahead forecasts
- Publication dates of the weights from the Hugging Face model repositories; they fix what the models could have seen
- Only Chronos-2 reaches the 1% level needed for VaR 1%

Pretraining Leakage

- A test is out of sample only if the test period was never seen during training
 - ▶ The Chronos corpus includes financial series (e.g. daily exchange rates 1990–2016); later corpora are broader
 - ▶ We cannot rule out that our series, or series driven by the same shocks, were in the pretraining data
- Our remedy: a **post-release window** from 3 November 2025, after all three sets of weights were published
 - ▶ Clean, but short (about 220 trading days; 320 calendar days for Bitcoin): low power, especially for VaR 1%
- Look-ahead in our own pipeline is avoided too: every scaling, window and estimate uses only past data

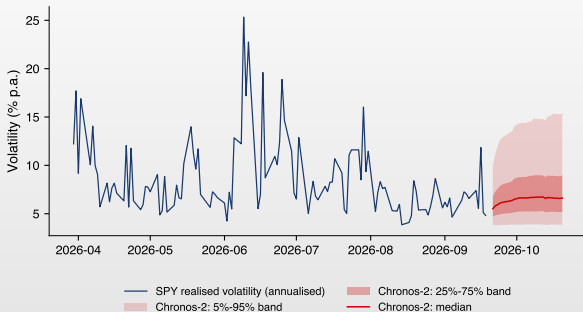
From a Quantile Grid to What We Need

- The model returns quantiles q_u at levels $u_1 < \dots < u_K$
 - ▶ Point forecast: the median $q_{0.5}$, or the mean $\int_0^1 q_u du$ approximated on the grid
 - ▶ Mean of a transformed variable: $E[g(Y)] = \int_0^1 g(q_u) du$ for increasing g (e.g. $g = \exp$ for log variance)
- Tail risk (Chapter 7): $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
 - ▶ Below the lowest level the grid is silent: we hold q_u at its last value, which understates ES
 - ▶ Chronos-Bolt and TimesFM stop at 10%: a "VaR 1%" request returns the 10% quantile

Worked Example: the Last Chronos-2 Forecast

- S&P 500, forecast for the day after 18.09.2026: $q_{0.01} = -2.70\%$, $q_{0.05} = -1.40\%$
 - ▶ VaR 1% = $-q_{0.01} = 2.70\%$
 - ▶ Linear interpolation: $q_{0.025} = q_{0.01} + \frac{0.015}{0.04}(q_{0.05} - q_{0.01}) = -2.21\%$, so VaR 2.5% = 2.21%
- ES 2.5%: $-\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.56\%$
 - ▶ A lower bound: the tail below 1% is cut at $q_{0.01}$

A Zero-Shot Fan Chart for Volatility



- Chronos-2 on log realised variance of SPY (512-day context), 22 trading days ahead from 18.09.2026; annualised volatility
- Bands: quantiles 5%–95% and 25%–75% of the predictive distribution

Interpretation: Mean Reversion Learned Without Training

- Last observed volatility 4.8% p.a.; average of the last 20 days 6.6%
 - ▶ Median forecast at 22 days: 6.6%, with a 90% band from 3.8% to 15.4%
- The band widens with the horizon and is skewed upwards: volatility can jump, not collapse
 - ▶ Features of GARCH and HAR forecasts (Chapters 5 and 9), obtained here with no estimation on SPY data
- Whether these forecasts are **better** is an empirical question: next sections

Section Summary

- ▣ Foundation models: pretrained once on large corpora, used zero-shot with a context window
- ▣ They read a series through scaling, tokens or patches, and return quantiles
- ▣ Quantile grids limit tail estimates; pretraining overlap requires a post-release test window

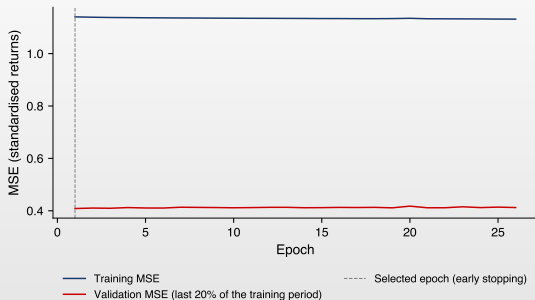
Design of the Return Test

- Daily log returns (%): S&P 500 and BET from 04.01.2016 (2 693 and 2 675 days), Bitcoin from 01.01.2019 (2 818 days)
 - ▶ Every forecast uses only data up to the previous close
- Models: historical mean, AR(1) on a 1,000-day window, LSTM, Chronos-2 and Chronos-Bolt (median), TimesFM-2.5 (mean)
- Benchmark: the zero forecast; out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$
 - ▶ Zero rather than the historical mean, as in [Gu, Kelly & Xiu \(2020\)](#); compare [Campbell & Thompson \(2008\)](#)
 - ▶ Inference: block-bootstrap CI for R^2 (blocks of 20 days), Diebold–Mariano (DM) test on squared errors with Newey–West variance ([Diebold & Mariano \(1995\)](#))

The LSTM for Returns

- ▣ Input: the last 20 returns, standardised with the mean and standard deviation of the training period
 - ▶ Output: next-day return; loss MSE; Adam; weight decay 10^{-4}
- ▣ Trained once on up to 4,000 days before the test period; last 20% of them for early stopping
 - ▶ Five random seeds; the forecast is the average of the five networks
- ▣ One hidden layer of 16 units: 1 233 parameters

Training the LSTM: Early Stopping

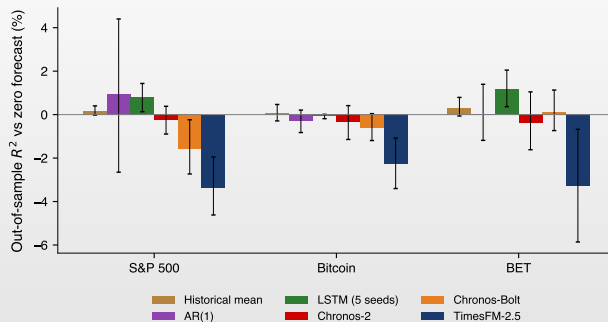


- S&P 500, seed 0: MSE of standardised returns by epoch, on the training and validation parts
- Training stops 26 epochs in; the weights of epoch 1 are kept

Interpretation: Little to Learn

- Both losses barely move; the validation loss is lowest after the first epoch, so early stopping keeps epoch 1
 - ▶ The validation MSE is below the training MSE because the validation years were calmer: the level reflects volatility, not skill
- Early stopping picks a network close to its starting point: forecasts near the mean
 - ▶ Standard deviation of the LSTM forecasts for the S&P 500: 0.047%, against 1.13% for returns
- A healthy sign: a network that forecast large moves would be fitting noise

Out-of-Sample R^2 for Returns



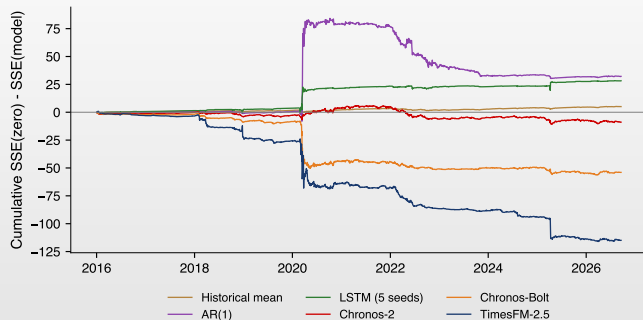
□ R^2 against the zero forecast (%), with 95% block-bootstrap CIs

Interpretation: No Model Beats Zero, Except on the BET

R^2 (%) [DM p]	Historical mean	AR(1)	LSTM (5 seeds)	Chronos-2	Chronos-Bolt	TimesFM-2.5
S&P 500	+0.15 [0.095]	+0.94 [0.639]	+0.82 [0.116]	-0.26 [0.418]	-1.57 [0.078]	-3.35 [0.001]
Bitcoin	+0.07 [0.691]	-0.28 [0.326]	-0.08 [0.135]	-0.35 [0.413]	-0.61 [0.070]	-2.29 [0.001]
BET	+0.30 [0.081]	-0.00 [1.000]	+1.18 [0.015]	-0.39 [0.562]	+0.12 [0.831]	-3.27 [0.070]

- S&P 500 and Bitcoin: no model beats zero significantly; TimesFM-2.5 is significantly **worse** than zero
 - ▶ Foundation models forecast too much movement: the s.d. of TimesFM forecasts on the S&P 500 is 0.141%, three times that of the LSTM (0.047%)
- BET: LSTM R^2 +1.18% (DM $p = 0.015$): a small but real signal in a less efficient market (Chapter 2)

Cumulative Squared-Error Gains on the S&P 500

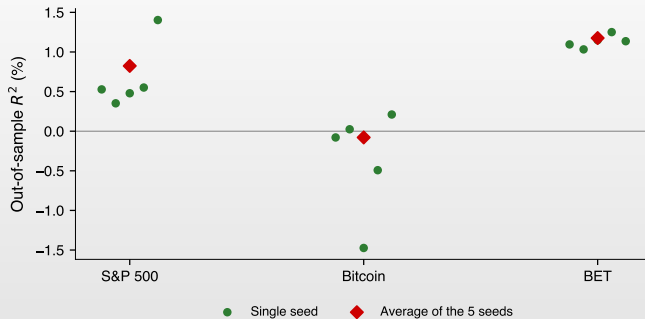


□ $\sum_{s \leq t} [r_s^2 - (r_s - \hat{r}_s)^2]$: rises when the model beats the zero forecast, falls when it loses

Interpretation: Gains Come from a Few Days

- All curves stay near zero in calm years and jump in March 2020
 - ▶ Squared errors are dominated by crisis days: one month decides the ranking
- AR(1) gains in 2020 from the negative autocorrelation of crash days, then gives part back
 - ▶ TimesFM-2.5 loses steadily: its forecasts are too volatile
- Lesson: report the path, not only the final R^2 ; a single episode can create or destroy "predictability"

The Seed Lottery



- Out-of-sample R^2 of five LSTMs that differ only in the random seed, and of their average

Interpretation: Report All Seeds

- ▣ Bitcoin: single seeds range from -1.47% to +0.21%; S&P 500 from +0.35% to +1.40%
 - ▶ Seed-to-seed variation is as large as the effect itself
- ▣ BET: all 5 seeds positive, between +1.03% and +1.25%: the one robust signal
- ▣ Choosing the best seed after looking at test results is backtest overfitting (Chapter 13)

Section Summary

- ▣ Daily returns of large markets: no deep or foundation model beats the zero forecast
- ▣ Zero-shot point forecasts can be worse than zero: too much movement is predicted
- ▣ A small LSTM finds a robust signal in the BET, consistent with its lower efficiency

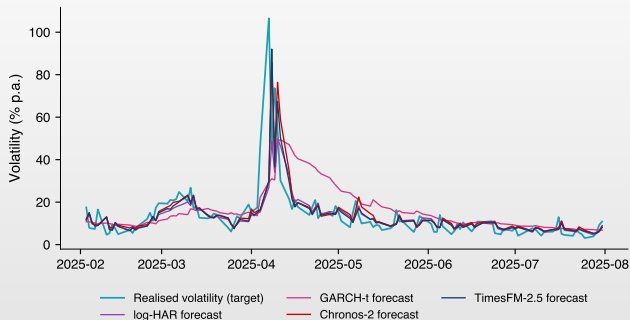
Target, Benchmark and Loss

- ▣ **Realised variance** (RV, [Andersen et al. \(2003\)](#)): sum of squared 5-minute log returns within the trading day
 - ▶ $RV_t = \sum_{j=1}^{78} r_{t,j}^2$ for SPY (regular session, 6.5 hours); average volatility 11.9% p.a.
- ▣ **HAR** ([Corsi \(2009\)](#)): $RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-5:t-1} + \beta_m \overline{RV}_{t-22:t-1} + \varepsilon_t$
 - ▶ log-HAR: the same on log RV, forecast $\exp(\hat{y} + \hat{s}^2/2)$
- ▣ **QLIKE** loss ([Patton \(2011\)](#)): $L = RV_t/\hat{h}_t - \log(RV_t/\hat{h}_t) - 1$
 - ▶ Ranks forecasts correctly even though RV is a noisy proxy; minimised by the conditional mean

Design of the Volatility Test

- ▣ SPY realised variance, one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days)
 - ▶ HAR and log-HAR on a rolling 500-day window; random walk: $\hat{h}_t = \text{RV}_{t-1}$
 - ▶ GARCH-t on daily SPY returns, rescaled to RV units by the ratio of average RV to average squared return over 500 days
 - ▶ LSTM on the last 22 values of log RV, retrained every 125 days on 500 days (hidden size 8, three seeds)
- ▣ Foundation models on log RV (context 512): forecast $\hat{h}_t = \int_0^1 \exp(q_u) du$ on the quantile grid
 - ▶ Post-release window: 218 days from 3 November 2025

Forecasts Through the April 2025 Shock

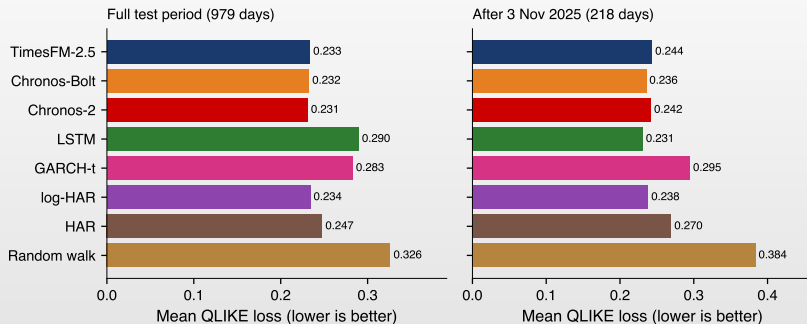


- SPY realised volatility (annualised) and one-day-ahead forecasts, February–July 2025
- Peak: 107% p.a. on 07.04.2025, after the US tariff announcements

Interpretation: Everybody Is One Day Late

- No model anticipates the jump; all of them react the next day
 - ▶ One-day forecasts can only extrapolate the past; shocks come from news
- The foundation models and log-HAR decay back at similar speeds
 - ▶ GARCH-t reacts to the size of daily returns and decays more slowly
- The ranking is decided by the average over about four years: next slide

QLIKE Losses of Eight Volatility Models



Left: full test period; right: window after the release of all model weights

Interpretation: Foundation Models Match log-HAR, Not More

Model	QLIKE	DM vs log-HAR	p	MCS p	QLIKE after 3.11.2025
Random walk	0.326	+4.53	<0.001	0.00	0.384
HAR	0.247	+1.30	0.194	0.26	0.270
log-HAR	0.234	—	—	0.95	0.238
GARCH-t	0.283	+3.43	<0.001	0.03	0.295
LSTM	0.290	+1.87	0.062	0.19	0.231
Chronos-2	0.231	-0.62	0.537	1.00	0.242
Chronos-Bolt	0.232	-0.18	0.857	0.95	0.236
TimesFM-2.5	0.233	-0.08	0.938	0.95	0.244

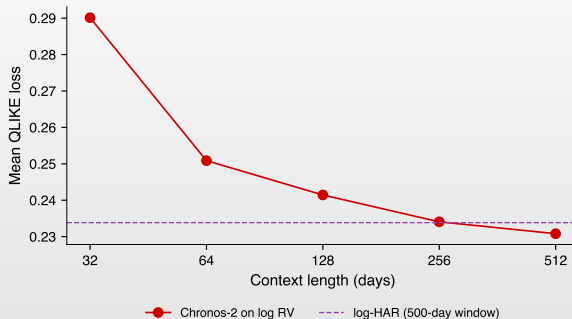
- Chronos-2, Chronos-Bolt, TimesFM-2.5 and log-HAR: DM tests not significant, all in the 90% MCS
- Random walk and GARCH-t are eliminated from the MCS; HAR in levels over-forecasts (mean forecast / mean RV = 1.36)
- After 3 November 2025 the small LSTM has the lowest QLIKE (0.231): over 218 days, rankings are unstable

Median or Mean? A Costly Detail

Chronos-2 and TimesFM-2.5 on log RV	QLIKE	Mean forecast / mean RV
Chronos-2, $\int_0^1 \exp(q_u) du$	0.231	1.02
Chronos-2, $\exp(q_{0.5})$	0.259	0.79
TimesFM-2.5, $\int_0^1 \exp(q_u) du$	0.233	0.94
TimesFM-2.5, $\exp(q_{0.5})$	0.259	0.79

- RV is right-skewed: its median is about 20% below its mean
 - ▶ $\exp(\text{median of log RV})$ is the median of RV, not its mean: a biased volatility forecast
 - ▶ The median version is significantly worse than the mean version (DM +3.60 for Chronos-2) and worse than log-HAR (0.234)

How Much Context Does Chronos-2 Need?



□ QLIKE of Chronos-2 on log RV with contexts of 32 to 512 days (same forecast days)

 MFM_ch14_realised_vol

Interpretation: Long Memory Needs Long Context

- ▣ QLIKE falls from 0.290 with 32 days to 0.231 with 512 days
 - ▶ With less than about 256 days, Chronos-2 is worse than log-HAR (0.234)
- ▣ Volatility has long memory: months of history tell the model the slow-moving level
 - ▶ HAR encodes the same idea explicitly, through its 22-day average
- ▣ Context length is a modelling choice to report and to test, like a window length

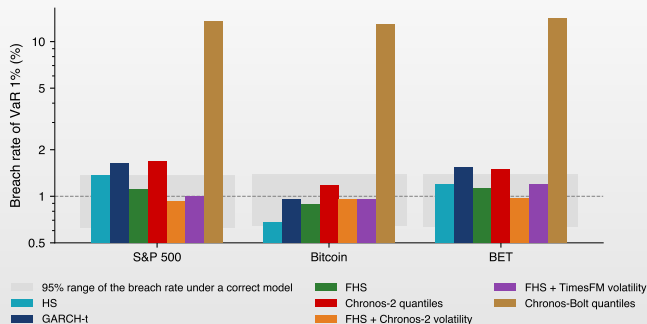
Section Summary

- ▣ Zero-shot foundation models forecast realised variance as well as a well-specified log-HAR
- ▣ They are not significantly better, before or after the release of their weights
- ▣ Use the mean, not the median, and a long context; always compare with HAR

Design of the Tail-Risk Test

- $\text{VaR}_\alpha = -q_\alpha(r_{t+1} \mid \mathcal{F}_t)$, the loss exceeded with probability α ; $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
 - ▶ Levels: VaR 1% (coverage tests) and the pair (VaR 2.5%, ES 2.5%) for the FZ0 loss (Fissler & Ziegel (2016), Patton, Ziegel & Chen (2019))
- Classical models on a 1,000-day window: HS, GARCH(1,1)-t, FHS (McNeil & Frey (2000)), re-estimated every 20 days
- Foundation models, two ways
 - ▶ **Raw**: Chronos-2 quantiles of tomorrow's return (context: 512 returns)
 - ▶ **Hybrid**: volatility $\hat{\sigma}_t$ = forecast mean of $|r_t|$ from Chronos-2 or TimesFM; FHS on $r_t/\hat{\sigma}_t$

VaR 1% Breach Rates



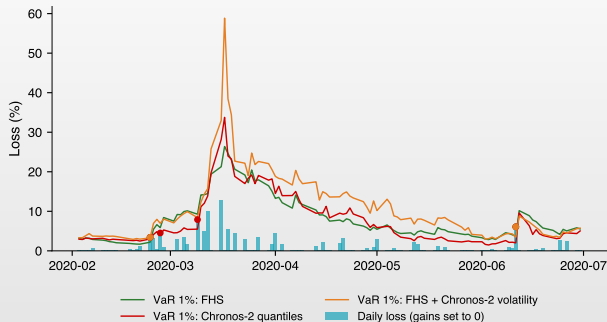
- Share of days with a loss above VaR 1% (log scale); grey band: 95% range of the breach rate if the model were correct
- Chronos-Bolt: its lowest quantile (10%) used as "VaR 1%"

Interpretation: Raw Quantiles Are Too Narrow

Breaches [Kupiec p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (26.9)	37 [0.065]	44 [0.002]	30 [0.559]	45 [0.001]	25 [0.705]	27 [0.989]
Bitcoin (28.2)	19 [0.065]	27 [0.822]	25 [0.539]	33 [0.374]	27 [0.822]	27 [0.822]
BET (26.8)	32 [0.322]	41 [0.010]	30 [0.536]	40 [0.016]	26 [0.884]	32 [0.322]

- In brackets after the market: the expected number of breaches, 1% of the days
- Raw Chronos-2 quantiles: too many breaches on the S&P 500 (1.67%); Chronos-Bolt "VaR 1%": 13.5%
 - ▶ Hybrid FHS with foundation-model volatility: breach rates close to 1% in all three markets

The COVID-19 Crash: VaR 1% of Three Models

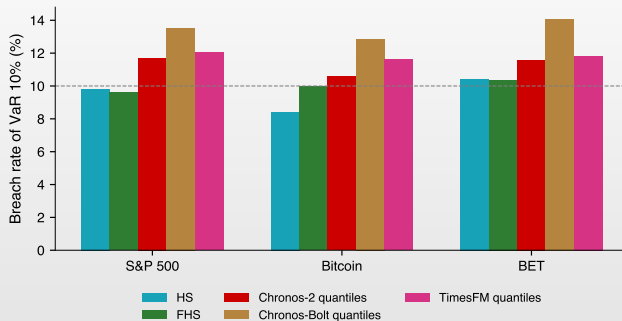


□ S&P 500, February–June 2020 (104 days): daily losses and VaR 1% forecasts; dots: breaches

Interpretation: Speed of Reaction

- Breaches in the 104 days: FHS 2, raw Chronos-2 4, FHS with Chronos-2 volatility 2, HS 12
 - ▶ All models are hit in the first days of the crash: none anticipates it
- Chronos-2 reads the context in the first days, so its quantiles widen fast
 - ▶ Scaling by the mean absolute value of the context: one crash day widens the whole distribution
- The hybrid overshoots: its VaR 1% peaks at 58.9%, against 26.4% for FHS and a largest loss of 12.8%
 - ▶ Few breaches, but a high capital cost: this is what the FZ0 loss will penalise

Where Deciles Suffice: VaR 10%



- Breach rate of the 10% quantile (target 10%) for HS, FHS and the three foundation models

Interpretation: Even the 10% Quantile Is Too High

- ▣ S&P 500: HS 9.8%, FHS 9.6%; Chronos-2 11.7%, Chronos-Bolt 13.5%, TimesFM-2.5 12.0%
 - ▶ Bitcoin: Chronos-2 10.6%, Chronos-Bolt 12.8%, TimesFM-2.5 11.6%; BET: 11.6%, 14.1%, 11.8%
- ▣ All three foundation models breach their 10% quantile too often: predictive distributions too narrow for returns
 - ▶ A plausible reason: pretraining corpora are dominated by smoother series (energy, traffic, web) with lighter tails than returns
- ▣ Used as volatility filters, the same models inherit the tails from the data, through FHS

Comparing (VaR 2.5%, ES 2.5%) Forecasts: the MCS

HS	0.140	0.343	0.015
GARCH-t	0.187	0.597	0.341
FHS	1.000	0.597	1.000
Chronos-2 quantiles	0.140	0.641	0.239
FHS + Chronos-2 volatility	0.140	1.000	0.341
FHS + TimesFM volatility	0.183	0.704	0.202
	S&P 500	Bitcoin	BET

□ p -values of the 90% model confidence set ([Hansen, Lunde & Nason \(2011\)](#)) for the FZ0 loss; green: kept, red: eliminated

Interpretation: Hybrids Pass, FHS Remains the Reference

FZ0 [MCS p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500	1.309 [0.14]	1.026 [0.19]	0.999 [1.00]	1.147 [0.14]	1.084 [0.14]	1.059 [0.18]
Bitcoin	2.237 [0.34]	2.192 [0.60]	2.192 [0.60]	2.181 [0.64]	2.151 [1.00]	2.160 [0.70]
BET	1.230 [0.01]	1.001 [0.34]	0.963 [1.00]	1.032 [0.24]	1.007 [0.34]	1.027 [0.20]

- S&P 500: FHS has the lowest FZ0 loss; hybrids cover correctly but with larger, more conservative VaR (3.20% vs 2.78% on average)
- Bitcoin: the hybrid FHS with Chronos-2 volatility has the lowest FZ0 loss, but the differences are not significant
- The MCS eliminates no model on the S&P 500 and Bitcoin, only HS on the BET; pairwise DM tests are sharper: FHS beats the Chronos-2 hybrid on the S&P 500 (DM +2.65, $p = 0.008$) and on the BET (DM +2.36, $p = 0.018$)

After the Release of the Weights

Breaches after 3.11.2025	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (220 days)	0	2	2	2	2	2
Bitcoin (320 days)	3	3	4	4	3	4
BET (217 days)	4	4	4	5	1	2

- Expected number of breaches: 2.2 (S&P 500), 3.2 (Bitcoin), 2.2 (BET)
 - ▶ No leakage is possible here, but with two or three expected breaches no test has power
- For tail risk, the post-release window cannot confirm or reject anything yet: it must grow for years

Section Summary

- ▣ Raw foundation-model quantiles are too narrow for returns, even at 10%; Chronos-Bolt and TimesFM cannot reach 1% at all
- ▣ As volatility filters inside FHS they give correct coverage, but a significantly higher FZ0 loss than GARCH-based FHS on the S&P 500 and BET
- ▣ None of them replaces GARCH-based FHS on the evidence so far

Pitfalls Checklist

- ▣ **Leakage:** pretraining overlap, scaling with full-sample statistics, shuffled validation windows
 - ▶ Seminar B8: HAR averages that include the forecast day cut QLIKE by a quarter, a gain no honest model achieves
- ▣ **Weak benchmarks:** compare with zero (returns), log-HAR (volatility), FHS (tails), not with a random walk
- ▣ **Selection:** seeds, context lengths, architectures chosen after seeing the test period
- ▣ **Wrong functional:** medians used as means; deciles used as 1% quantiles

When to Use What

Target	Reference model	Where deep learning can help
Daily index returns	Zero forecast	Less efficient markets (BET); many assets, many features
Realised volatility	log-HAR	Zero-shot forecasts for thousands of series without estimation
VaR 1%, ES 2.5%	FHS (GARCH filter)	Volatility filter inside FHS, not raw quantiles

- Foundation models are cheap to try: no estimation, one line of code, a CPU is enough
- Their value is breadth (many series, no tuning), not accuracy on a single well-studied series

Related Research

- ▣ Agakishiev, Härdle & Pele (2025): forecasting cryptocurrency liquidity, Springer Handbook of Computational Statistics
- ▣ Będowska-Sójka, Wójcik & Pele (2026): machine-learning early-warning systems for "zombie" crypto-assets
- ▣ Pele et al. (2023): cryptocurrencies classified with statistical learning across many series
- ▣ Textbooks: Franke, Härdle & Hafner (2019) and Härdle, Chen & Overbeck (2017) for the statistics of financial markets behind the benchmarks

Key Takeaways

- LSTMs solve the vanishing gradient through the forget gate; Transformers avoid it through attention
- Foundation models forecast zero-shot from a context window and return quantiles
- Returns: nothing beats zero on large markets; volatility: foundation models equal log-HAR; tails: use them only as volatility filters
- Evaluate every new model with the tools of Chapter 8, against a strong benchmark and after the release of its weights

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